NATIONAL TRANSPORTATION SAFETY BOARD
Investigative Hearing
US Army PAT25, Sikorsky UH-60L and
PSA Airlines flight 5342
MHI RJ Aviation CL-600-2C10 (CRJ700), N709PS
Midair Collision in Washington, DC
January 29, 2025

Docket No.	**SA-544**
EXHIBIT	
17-B	

Aviation Safety Reporting System (ASRS) Review and Analysis Process

(6 Pages)

National Transportation Safety Board
Office of Aviation Safety
Washington, DC 20594

DCA25MA108

SAFETY DATA

Group Chair's Factual Report – Attachment 1

Aviation Safety Reporting System (ASRS) Review and Analysis Process

June 18, 2025

Answers to Questions Submitted to the Aviation Safety Reporting System (ASRS) Program, received June 5, 2025.

1. How are ASRS reports vetted and processed?

Initial Screening: NASA's ASRS team, comprised of experienced aviation professionals each with over 10 years of operational experience in their domain (e.g. Airline pilot, ATC, Mx), screens every report within three to five days of receipt to identify hazards and vulnerabilities and to triage report processing. Critical safety issues identified in this stage trigger alerts to stakeholders (e.g., FAA, airlines, airport managers, manufacturers, industry groups) to determine if further action is warranted and for increased awareness of safety concerns to prevent similar events or incidents from becoming accidents.

Analysis and Coding: Analysts combine multiple reports about the same event (e.g. from Pilot Flying, Pilot Not Flying, ATC, etc) into a single record and apply a coding taxonomy which includes codes for anomaly type, contributing factors, primary problem, human factors, result or outcome, system component, event detector and more. This enables record retrieval from the ASRS database.

De-identification: The Identification Strip (containing submitter details) is detached, date-stamped, and returned to the submitter confirming receipt. Any information that can be used to identify the reporter or related third-parties (e.g., personal names, company names, dates, and flight numbers) are deidentified or generalized from the remaining fixed-fields and narrative on a report-by-report basis to preserve the important safety information while protecting reporter confidentiality. This allows for broad dissemination of important safety information without compromising reporter confidentiality.

Database Integration: De-identified reports are entered into the publicly available ASRS database for safety research and analyses by government agencies, aviation industry groups, academic and research organizations. This allows for the data to be broadly disseminated as opposed to remaining siloed within a single organization or agency.

The publicly available ASRS database is also distributed directly to the FAA (ASIAS) where the data are made broadly available to FAA researchers and operational specialists. This resource supports FAA's Safety Management System (SMS) and industry initiatives like the Commercial Aviation Safety Team (CAST). Many FAA offices like NextGen/ Wake Turbulence, Office of Hazardous Materials Safety, Airport Safety and Operations, Air Traffic Organization, Air Traffic Safety Standards use the data regularly. The data are also used by other government agencies (NTSB, DHS, DOD, NASA etc.), academia, industry groups (AOPA, ALPA, NBAA etc) and research organizations (e.g. MITRE, Volpe etc) who don't have access to internal FAA/ASIAS and ASAP/ATSAP reports.

2. Does someone read every report and then redact? Or is it only a subset of the total reports that get read and redacted?

   Every ASRS report is read (screened) by ASRS Expert Analysts within 3 to 5 days of receipt according to the Initial Screening process identified above for the purposes of issuing safety alerts and triaging report processing. Reports that are selected for further processing receive coding, and deidentification as described above.

3. Does anyone reach out to the submitter to get additional information?

   ASRS analysts may contact submitters if they require more information to understand the nature of the safety issue being reported or for action to be taken to investigate or resolve the safety issue. Any additional information offered by the reporter during the callback will be de-identified and entered into the record as a Callback Note.

4. Are trends identified, and if so, what happens with that information?

   ASRS maintains records of report intake volume by month and year, and tracks report intake by reporter function, event (anomaly) type, and other parameters. Annual summaries are presented in the program briefing on the ASRS website.

   The mission of ASRS is to make data widely available to other government agencies and industry to make sure that information does not stay siloed within one organization. Several FAA offices and ASIAS use ASRS reports to identify systemic issues throughout the NAS. The data are available for other government agencies (NTSB, DHS, DOD, NASA etc.), academia, operational and research organizations who frequently conduct analyses to answer operational, research, and policy-related questions. The database is mined to understand contributing factors that are used to identify and propose corrective actions.

   In addition to the ASRS database, ASRS disseminates safety information in the following ways:

   - Safety Alert Bulletins and For Your Information (FYI) notices are issued to the proper authority(s) to investigate the safety issue and determine if an action is warranted.
   - ASRS Safety Telecons with FAA and NTSB are held monthly to discuss complex and emerging issues.
   - Search Requests - upon request ASRS will generate data compilations using automated search terms and human expert review for relevance
   - Data Report sets – a curated collection of the 50 most recent reports pertaining to 30 specific topics is maintained and refreshed semi-annually

- Monthly issues of the CALLBACK newsletter and quarterly issues of the UAS Safety in Sight newsletter are published and used widely in aviation industry for training.
- Industry Meetings (e.g. FAA InfoShare, Business Aviation Safety Summit, FAA GASA, Drone Safety Team)
- Secondary analyses and Special Studies as requested to extract contributing factors and corrective actions on selected topics. Current and past topic area include: Wake Encounters following changes in Dynamic spacing procedures, HAZMAT, RNAV and Metroplex procedures, COVID, 5G interference, GPS interference, Boeing 737 Max, UAS/Drone airspace violations and near mid-air collisions with aircraft.

Common uses of ASRS safety reports and safety products include:

Identifying and correcting deficiencies and discrepancies in the NAS. Examples include: Charting errors, Airport Signage and Markings, taxiway/runway/ramp lighting, Similar Sounding Fixes, PAPI misalignment, glide slope anomaly

Identifying systemic aircraft, equipment, and automation failures and vulnerabilities: Examples include aileron control failure, recurring brake damage

Policy and ATC Procedure Assessment: Examples include airspace design, RNAV approach design, dynamic spacing, and NOTAM modernization.

Training: Examples include: human factors and crew resource management training for pilots and controllers, Pilot decision making for go-arounds, and pilot familiarization with 'tribal knowledge' when flying into new airports.

Supporting Incident and Accident Investigation. Following major aircraft accidents, or upon request by NTSB, ASRS compiles relevant data reports that may provide insight into the accident investigation.

Understanding Implications and Corrective Actions for Broad Complex Safety Issues. Examples include: crew resource management, fatigue and scheduling, automation management, crew proficiency, impact of airspace resectorization on Air Traffic Controllers.

Understanding Emerging Operational Threats. Examples include: integration of drones and e-VTOL vehicles into the NAS, impact of space launches or tourism ballons on aircraft operations, conflicts between general aviation and commercial aircraft at non-towered aircraft

Stakeholder Engagement: Data are shared with interagency partners and industry groups to drive systemic safety enhancements, reflecting federal mandates for data-driven decision-making.

5. What proportion of ASRS reports are made directly to the system versus referred through voluntary safety reporting systems such as ASAP and ATSAP?

This number can fluctuate, but NASA supplies monthly activity reports which supplies these numbers. Attached and below are the latest (April 2025) statistics:

Reporting Sources	April Intake	April Percentage	2025 Cumulative Counts	2025 Cumulative Percent
Direct/ERS	1,471	14.72%	5,508	14.12%
Direct/Paper	7	0.07%	53	0.14%
ASAP/ATSAP Electronic	8,516	85.21%	33,439	85.74%
ASAP/Paper	-	0.00%	-	0.00%
TOTAL	9,994	100%	39,000	100%

- ASRS received a total of 9,994 reports in the month of April.
    - ASRS received 1,471 reports through Electronic Report Submission (ERS) and 7 paper reports.
    - ASRS received 8,516 reports through the ASAP/ATSAP secure electronic report transfer submission protocols.

By agreement between FAA and NASA, the ASAP/ATSAP safety reports that fall within high-priority safety-critical categories (such as Runway Incursions, Near Mid Air Collision, and Airborne Loss of Control) are processed, de-identified, and posted to the publicly available database to ensure broad distribution of the critical safety information and ensure it does not stay siloed within an airline or FAA/ASIAS.